



Republic of Rwanda
Ministry of Education



SWDND501

NoSQL DATABASE DEVELOPMENT

Develop NoSQL Database

Competence

RQF Level: 5

Learning Hours



60

Credits: 6

Sector: ICT & MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: Specific

Curriculum: ICTSWD5001-TVET Certificate V in Software Development

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Purpose statement	This specific module describes the knowledge, skills and attitude required to Develop NoSQL databases. This module is intended to prepare students pursuing TVET Level 5 in Software Development. At the end of this module the student will be able to Prepare database environment, Design database, Implement database, Manage Database.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	Group project and presentation	20%				
	Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Elements of competence	Performance criteria
1.Prepare database environment	1.1 Database requirements are properly identified based on user requirements
	1.2 Database is clearly analysed based on database requirements.
	1.3 Database environment is successfully prepared based on established standards.
2.Design database	2.1 Drawing tools are properly selected based on database requirements.
	2.2 Conceptual Data Modeling is created based on the structure of the data and its relationships.
	2.3 Database Schema is clearly designed according to Mongoose.
3. Implement database design	3.1 MongoDB data definition are properly performed based on database requirements
	3.2 MongoDB data manipulation are properly performed based on database requirements
	3.3 Query optimizations are properly applied based on query performance.
4. Manage Database	4.1 Database users are effectively managed with appropriate permissions.
	4.2 Database is effectively secured in line with best practices.
	4.3 Database is successfully deployed based on the targeted environment.

Knowledge, Skills, and Attitude

Knowledge	Skills	Attitude
- Identifying Database requirements - Identifying user requirements - Identify the scalability of mongo DB	- Creation of use cases. - Performing Data analysis - Implementation of Data validation - Setting up MongoDB environment - Designing MongoDB Database Schema	- Curiosity - Problem-Solving Orientation - Attention to Detail - Continuous Learning - Adaptability - Team Collaboration

● Description of requirements analysis process ● Identification of noSQL drawing tools ● Identification of Collections ● Definition of Collection Structure ● Description of Optimization techniques ● Identification of the role of database users ● Identification of mongoDB Cluster Architectures	● Validation and Normalization of Schema ● Applying Design Patterns ● Performing MongoDB Data Definition ● Executing data manipulation ● Applying Mongosh Methods ● Evaluating Performance of Current Operations ● Optimizing Query Performance ● Creating Users ● Managing Roles and privilege ● Enabling Access Control and Enforce Authentication ● Configuring Role-Based Access Control ● Performing Backup and recovery ● Applying deployment Options ● Scaling MongoDB with Sharding	
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Learning outcomes	At the end of the module the learner will be able to: 1. Prepare database environment 2. Design database 3. Implement database 4. Manage Database
Learning outcome 1: Prepare database environment	Learning hours: 10
Indicative content	
● Identifying Database requirements ✓ Definition of key terms ✚ NoSQL ✚ MongoDB ✚ Availability ✚ Documents ✚ Collection ✚ Indexing ✚ Optimistic Locking ✚ Relationships ✚ Data model ✚ Schema ✚ Mongosh ✓ Identifying user requirements ✓ Describe: ✚ Characteristics of collections. ✚ Features of NoSQL Databases ✚ Types of NoSQL Databases ✚ Data Types	

✓ Defining use cases.

- **Analysing NoSQL Database**

✓ Describe requirements analysis process

- ✚ Identify Key Stakeholders and End-Users.

- ✚ Capture Requirements.

- ✚ Categorize Requirements.

- ✚ Interpret and Record Requirements.

- ✚ Validate Requirements

✓ Perform Data analysis

✓ Implement Data validation

- **Preparing Database environment.**

✓ identify the scalability of mongoDB

✓ Setting up MongoDB environment

- ✚ Shell environment

- ✚ Compass environment

- ✚ Atlas environment

Resources required for the learning outcome

Equipment	● Computer ● Projector
Materials	● Internet
Tools	● Mongosh ● MongoDB compass ● Browser ● MongoDB ● Apache Cassandra ● Redis ● Neo4j

Facilitation techniques	● Brainstorming ● Demonstration ● Group discussion practical work ● Trainer guided
Formative assessment methods	● Written assessment ● Oral presentation ● Performance assessment

Learning outcome 2: Design NoSQL database	Learning hours: 20
Indicative content	
● Selecting tools of drawing databases. ✓ Identify NoSQL drawing tools ✓ Installation of Edraw Max drawing tool ● Creating Conceptual Data Model. ✓ Identify Collections ✓ Model Entity Relationships ✓ Define sharding and replication ✓ Visualize High-Level Data Model  UML Class Diagrams  Data Flow Diagrams (DFDs) ✓ Design a conceptual data model ● Designing MongoDB Database Schema ✓ Identify Application Workload ✓ Define Collection Structure ✓ Map Schema Relationships ✓ Validate and Normalize Schema	

✓ Apply Design Patterns	
Resources required for the indicative content	
Equipment	• Computer
Materials	• Internet
Tools	• Edraw Max • Mongosh • MongoDB compass
Facilitation techniques or Learning activity	• Brainstorming • Group discussion • Demonstration • Practical exercise • Trainer guided
Formative assessment methods /(CAT)	• Written assessment • Oral presentation • Performance assessment

Learning outcome 3: Implement Database Design	Learning hours: 20
Indicative content	
• Perform MongoDB Data Definition ✓ Create  Database  collections ✓ Drop  Database	

 Collections

✓ Rename

 Database

 collections

- **MongoDB data Manipulating**

✓ Execute data manipulation

 Insert document

 Update

 Delete

 Replacing Documents

 Querying Documents

 Indexes

✓ Bulk Write Operations

✓ Aggregation Operations

✓ Apply Mongosh Methods

 Collection Methods

 Cursor methods

 Database Methods

 Query plan cache methods

 Bulk operation methods

 User management methods

 Role management methods

 Replication methods

 Sharding methods

 Free monitoring methods

 Object constructors and methods

 Connection methods

 Atlas search index methods

- **Apply Query optimizations**

✓ Describe Optimization techniques

✓ Evaluate Performance of Current Operations

✓ Optimize Query Performance

Resources required for the indicative content

Equipment	● Computer
Materials	● Internet
Tools	● MongoDB ● Mongosh ● MongoDB compass ● MongoDB Shell
Facilitation techniques or Learning activity	● Brainstorming ● Demonstration ● Group discussion practical work ● Individual practical work ● Trainer guided
Formative assessment methods /(CAT)	● Written assessment ● Oral presentation ● Performance assessment ● Project based assessment

Learning outcome 4: Manage Mongoddb database

Learning hours: 10

Indicative content

- **Management of Database users**
 - ✓ Identify the role of database users
 - ✓ Creating Users
 - ✓ Manage Roles and privilege
- **Securing Database**
 - ✓ Enable Access Control and Enforce Authentication
 - ✓ Configure Role-Based Access Control
 - ✓ Data Encryption and Protect Data
 - ✓ Audit System Activity
 - ✓ Perform Backup and Disaster Recovery
- **Deployment of Database**
 - ✓ Applying deployment Options
 - 🚧 On-Premises
 - 🚧 Cloud
 - 🚧 Hybrid
 - ✓ Identify mongoDB Cluster Architectures
 - 🚧 Single-Node
 - 🚧 Replica Set
 - 🚧 Sharded Cluster
 - ✓ Scaling MongoDB with Sharding

Resources required for the indicative content

Equipment	● Computer
Materials	● Internet
Tools	● Mongosh ● MongoDB compass ● MongoDB

Facilitation techniques or Learning activity	● Brainstorming ● Demonstration ● Group discussion practical work ● Individual practical work ● Trainer guided
Formative assessment methods /(CAT)	● Written assessment ● Oral presentation ● Performance assessment ● Project based assessment

Integrated/Summative assessment

MZR Ltd, a transportation company based in Rwanda with its primary operations centred in Kigali but extending services throughout the country, is facing challenges in efficiently managing ticket bookings and travel management. The existing manual system utilized for recording data is proving inadequate to meet the customer needs, leading to inefficiencies and operational hurdles.

MZR Ltd hired you as a database developer to implement a comprehensive mongoDB system aimed at streamlining ticket bookings and travel management. This database system should address the company's challenges in managing transportation services efficiently, particularly in the context of its operations in Rwanda.

Tasks to be performed:

1. Assess the limitations and inefficiencies of the existing manual system.
2. Design a comprehensive database schema tailored to MZR Ltd's specific needs.
3. Develop the database solution based on the design specifications.
4. Test the developed database system for functionality and reliability.

5. Deploy the database to MZR Ltd's server.

Instructions:

- Build the database with respect of collections and indexing
- No Data redundancy allowed in database
- MZR Ltd has the following employees' positions:

Managing director: They manage all company activities

Database administrator: manage all system activities

N.B: The above tasks should be completed in 8 Hours

All materials, equipment and tools will be provided by MZR Ltd

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Prepare database environment (20%)	Gathering requirements are properly identifying	Ind.1:Database requirements are Identified			6
		Ind.2:Database is analysed			6
		Ind.3:Database environment are Prepared			8
Learning outcome 2: Design database (30%)	1.Conceptual and logical database schema are properly designed	Ind.1:Drawing tools are selected			6
		Ind.2:Conceptual Data Model is created			11
		Ind.3:			13

		MongoDB Database Schema is Designed			
Learning outcome 3: Implement Mongoddb database (3/0%)	Mongoddb database is properly implemented	Ind.1: MongoDB and Collections are created			6
		Ind.2: insert , update search , view queries are performed			15
		Ind.3:Query optimizations are Applied			9
Learning outcome 4: Manage Database (20%)	Database security and user’s privilege are properly Applied	Ind.1: User’s data are encrypted			8
		Ind.2:Database users are created			5
		Ind.3:Database deployment is performed			7
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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